Materials and Stress Analysis

Convertible Wheelchair-to-Bed Transfer System

1. Materials used

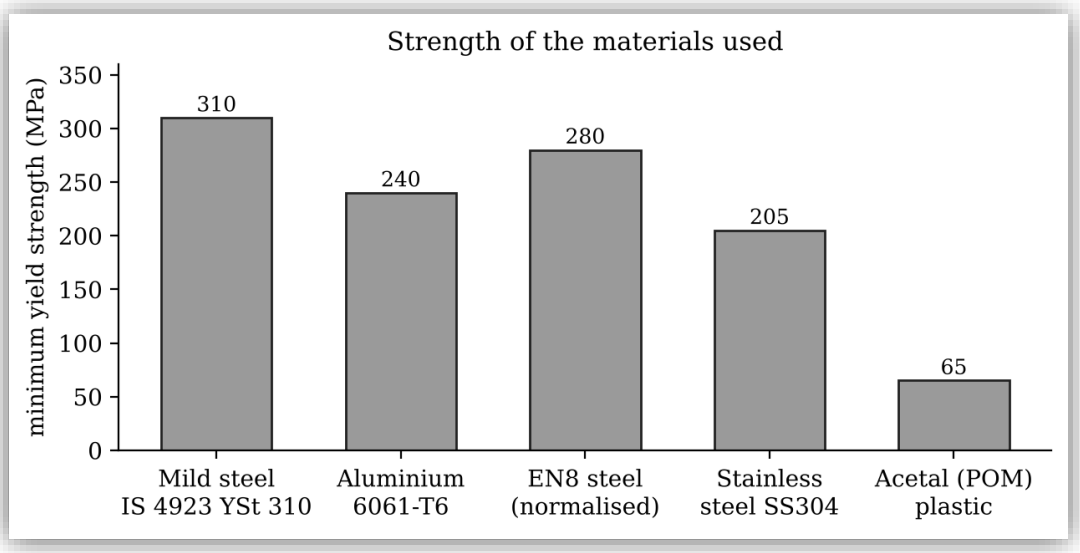
Parts	Material	Why this material
Base frame, scissor arms, lift top frame	Mild steel tubes (IS 4923, grade YSt 310)	These carry the heaviest loads in the chair. Steel is stiff, cheap and easy to weld. Its weight sits low down, which makes the chair harder to tip.
Seat, backrest and leg rest frames, side plates	Aluminium (6061-T6)	These parts move every time the chair reclines. Aluminium weighs about a third as much as steel, so the gas spring and the caregiver do less work. The parts are bolted, so no aluminium welding is needed.
Sliding boards	Aluminium tubes under a thin aluminium sheet, with slippery PTFE tape on top	The boards must be thin, light and stiff. The slippery top lets the belt slide over the board easily.
Pins, lead screw, roller axles	EN8 steel	Stronger than mild steel and easy to machine on a lathe. Can be hardened later if the pins wear.
Recline linkage	Mild steel plate	The pins press hard on their holes. Steel holes keep their shape better than aluminium ones.
Bed clamp	Mild steel, powder coated; steel screws	The clamp has the highest local loads among the safety parts. Steel threads survive being tightened again and again.
Lock posts, edge bars, stop blocks	Aluminium, machined	The loads here are small, and these parts bolt onto aluminium frames.
Locking pins (plungers)	Stainless steel (SS304) or hardened steel	Bought as standard parts; stainless steel does not rust.
Handles and armrests	Mild steel tube, powder coated	Stiff, cheap, easy to bend and comfortable to hold.
Rack and pinion	Acetal (POM) plastic	The forces are small. Acetal is quiet, needs no grease near the patient, and does not rust.
Belts	Polyester covered with PU	The polyester barely stretches, so the belt keeps its tension. The PU surface is smooth and easy to wipe clean.

2. Strength of the Materials

The table gives the strength values used in every safety factor in this document. We used the **guaranteed minimum** values from the material standards, not the typical values. Real material is usually a little stronger, so the real safety factors are slightly higher than the ones shown.

Material	Yield strength (bends permanently)	Tensile strength (breaks)	Shear yield strength (pins)	Stiffness E	Density
Mild steel IS 4923 YSt 310	310 MPa	450 MPa	179 MPa	200 GPa	7.85 g/cm³
Aluminium 6061-T6	240 MPa (typical 276)	260 MPa (typical 310)	138 MPa	69 GPa	2.70 g/cm³
EN8 steel (080M40), normalised	280 MPa	550 MPa	162 MPa	205 GPa	7.85 g/cm³
Stainless steel SS304	205 MPa	515 MPa	118 MPa	193 GPa	8.0 g/cm³
Acetal (POM)	about 65 MPa	about 65 MPa	-	2.8 GPa	1.41 g/cm³

- **Safety factor = material strength ÷ stress in the part.**
- For parts that are bent or stretched (tubes, plates, boards), the yield strength is used.
- For pins that could be cut through, the shear yield strength is used. It is about 0.58 times the yield strength.
- For the plastic gear teeth, a lower working limit of 35 MPa is used, because plastic weakens under repeated loading.
- For the long, thin tie-rod, the buckling load is used instead of the material strength.



3. Dimensions of the main load-carrying parts

I (how stiff the cross-section is in bending) and **Z** (how much bending it can take as bending stress = bending moment ÷ Z).

Part	Material	Cross-section (mm)	Length (mm)	Area (mm^2)	I (mm^4)	Z (mm^3)
Scissor arms (×4)	Mild steel	Tube 50 × 25 × 2.6 wall	550 between pins	363	112,056	4,482
Base frame	Mild steel	Tube 40 × 40 × 2.6 wall	1040 × 700 frame	389	91,115	4,556
Lift top frame, seat side rails	Mild steel / aluminium	Tube 40 × 20 × 2 wall	630 / 517	224	44,459	2,223
Backrest side tubes (×2)	Aluminium	Tube 40 × 20 × 3 wall	818	324	60,812	3,041
Scissor cross shafts (×4)	Mild steel	Round tube Ø25 × 3 wall	424	207	12,778	1,022
Lead screw	EN8 steel	Tr20 × 4 thread, core Ø15.5	780	189	-	-
Tie-rod	Mild steel	Flat bar 25 × 8	482 between pins	200	1,067 (thin way)	267
Sliding board (back)	Aluminium	5 tubes 25 × 16.5 × 1.5 + 1.3 sheet, 17.8 deep	760	1,207	48,180	3,738
Bed clamp arm (×2)	Mild steel	Tube 25 × 25 × 2 wall	90	180	14,800	1,233
Pins	EN8 / SS304	Centre pin Ø20, pivot pins Ø12, locking pins Ø8	-	314 / 113 / 50	-	-

4. Loads that the chair can handle

The patient

The chair is designed for a patient of up to **120 kg**. All checks use 1.5 times this load (about 180 kg). For the seat, where a patient may sit down hard, the load is increased by a further 1.3 times.

How a lying patient's weight is shared

Part of the chair	Body parts resting on it	Share of body weight	For a 120 kg patient
Backrest	Head, chest, arms and stomach	49%	about 59 kg
Seat	Hips and thighs	39%	about 46 kg
Leg rest	Lower legs and feet	12%	about 15 kg

The chair itself and people leaning on it

The chair weighs about **89 kg**. The checks also include someone leaning on the head end of the flat chair with about 25 kg of force, and the patient leaning back hard against the backrest when sitting upright. Each part is checked in its **worst position**, for example the lift at its lowest height or the backrest fully flat. If a part is safe there, it is safe everywhere else.

5. Strength Checks

Part	Worst case checked	Highest stress (FEA)	Material strength	Safety factor
Scissor arms	Lift at its lowest height, chair flat	102 MPa (next to the centre pin hole)	310 MPa	3.0
Backrest side tubes	Chair flat, someone leaning on the head end	89 MPa at the hinge	240 MPa	2.7
Backrest sliding board	Upper body on the board plus 250 N in the middle	35.4 MPa	240 MPa	6.8
Bed clamp arm	Holding the chair at the worst moment of a transfer	34.5 MPa	310 MPa	9.0
Tie-rod (leg rest link)	Pushed from both ends (buckling)	Buckles at 9.06 kN; carries 831 N	-	10.9

Deflection	FEA	Hand calculation
Backrest side tube, bend at 500 mm from the hinge	5.42 mm	5.38 mm
Backrest sliding board, sag in the middle	2.27 mm	2.23 mm

Other parts: The smaller parts were checked by hand calculation **using AI**:

Part	What could go wrong	Stress or load	Strength used	Safety factor
Scissor centre pin	Cut through	18.3 MPa	162 MPa (shear)	8.8
Holes at the arm ends	Pushed out of shape by the shafts	44.0 MPa	310 MPa	7.1
Lead screw	Snaps under the pull of the lift	60.5 MPa	280 MPa	4.6
Backrest pivot pin	Cut through	20.2 MPa	162 MPa (shear)	8.0
Stop blocks (flat position)	Crushed	9.8 MPa	240 MPa	25
Backrest locking pin	Cut through if the gas spring fails while the patient leans back hard	36.9 MPa	118 MPa (shear)	3.2
Seat board	Bends when the patient sits down hard	50.8 MPa	240 MPa	4.7
Rack and pinion teeth	A plastic tooth breaks (the belt stops; the patient stays put)	15.3 MPa	35 MPa	2.3
Board lock pin	Cut through by a knock while moving	11.7 MPa	118 MPa (shear)	10.1
Bed clamp pivot pin	Bends during a transfer	54.9 MPa	310 MPa	5.6
Clamp mounting bolts (M6)	Pull out	719 N each	12,000 N	16.7

6. Effort for the caregiver

Task	Force needed	Feels like lifting
Turning the crank, lowest height (120 kg patient)	115 N	12 kg
Turning the crank at 55 cm	79 N	8 kg
Turning the crank at 65 cm	46 N	5 kg
Reclining the backrest (empty or with a 120 kg patient)	at most 76 N	8 kg
Sliding a 70 kg patient onto the bed	80 N	8 kg
Sliding a 120 kg patient onto the bed	137 N	14 kg
Moving a seated patient onto a commode	114 N	12 kg

The lift cannot sink by itself: the screw thread is so shallow (4.0°) that friction (equal to 7.1°) holds it in place.

7. Stability and anchoring to the bed

Position (120 kg patient on board)	Forward	Backward	Sideways
Chair, lowest height (50 cm)	39°	31°	27°
Chair, highest height (78 cm)	31°	23°	20°
Flat, highest height (78 cm)	41°	20°	23°

The table shows how far the chair could be tilted before it tips. Even in the worst case it is about 20°; a wheelchair ramp slopes only about 5°.

During a transfer the patient is carried out over the bed, past the chair's own wheels. At the worst moment the patient's weight tries to tip the chair with 294 N·m, while the chair's own weight resists with only 253 N·m. Each clamp needs to hold about 436 N (44 kg); a clamp screw tightened by hand gives about 3.3 kN, about 7.6 times more.

8. Assumptions and limits

- The chair is rated for a 120 kg patient. Heavier patients are outside the rating.
- Body weight sharing is based on standard data for a 1.75 m adult. Taller or shorter patients change the loads slightly, and the safety factors cover this.
- Material strengths are the guaranteed minimum values. Friction values are typical values for clean, dry parts and will be measured on the prototype.
- The FEA studies are linear and static: they assume the material stays below its yield strength (which the results confirm) and the load is applied slowly. Pins are modelled as rigid supports, which slightly over-estimates the stress right at the holes.
- Each FEA study models one part with simplified supports, not the whole chair at once. The supports were chosen to reproduce the worst load each part sees.
- Long-term wear from repeated use was not calculated in detail, but the everyday stresses are well below the limits for repeated loading.
- The chair weighs about 89 kg, above the 70 kg target. The parts with large safety factors show where weight can be removed in the future models of the chair.

AI was used in some places (to find angles, cross check stress calculations, etc)